



New battery breakthrough?

Graphene a-ion batteries reportedly charge 60% faster and are more sustainable than li-ion batteries. <http://digit.in/jun21-57>



Thoughts-to-text

An experimental device allowed a paralyzed patient to translate thoughts into text by imagining writing. <http://digit.in/jun21-58>



HOW DINOS WALKED

The good news is that you can easily outrun a T-Rex

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ow dinosaurs walked has been one of the pressing questions in dino science, and historically it has

been a very challenging task because of multiple reasons. The primary source of information for dinosaurs are the fossilised bones themselves, but palaeontologists have to figure out how these bones fit together in a complete animal, and make guesses about how they moved based on other similar animals. The second source of information are fossilised tracks, but these have to be matched with the creatures known from the fossil bones. At times experts can misiden-

tify the animal, for example 120 million year old footprints from South Korea, which was believed to be a pterosaur, actually turned out to be a bipedal crocodile.

Over time, scientists have suggested all kinds of ridiculous gaits for dinosaurs, including brontosaurus that dragged their bellies through the mud leaving a trail, dinosaurs that could only walk when supported by water, or dinosaurs that had to permanently keep their noses close to the ground. All of these are outdated ideas, and now researchers are using computer modelling, along with data mined from multiple specimens to make increasingly accurate guesses of the gaits used by dinosaurs, their stance, and how fast they can walk. Scientists can measure the distance between the prints in a fossilised track, and compare them to similar gaps between prints made of running birds, to figure out how fast the dinosaur was moving. This is still something that we can only guess

at, and there are so many different ways in which all the data from the fossil prints, actual specimens, and living organisms are churned. It is a constantly improving process, that researchers can only tackle the question by putting increasing limits on say how fast a particular dinosaur could run, how high it could jump, and find out characteristics about their gaits. For example we now know that if a T-Rex ran at more than about 20 kilometers per hour, then their bones would just explode.

Humans, animals, and birds all have a preferred walking speed, which is simply the least amount of energy you need to walk. In order to hit this peak efficiency, the timings of the swings of the limbs have to be coordinated just right to maintain a comfortable, relaxing pace. From cats, to humans to ostriches, all animals achieve what is known as a resonance at the walking speed optimised for energy efficiency. The whole body is in resonance at this time, which